

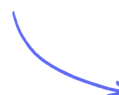
IGCSE · Cambridge (CIE) · Maths

 25 mins  6 questions

Investigation: Division by 11

Investigations

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Total Marks

/25

- 1 This investigation looks at numbers that are divisible by 11, in particular exploring the patterns in the digits of these numbers.

Two-digit numbers that are divisible by 11 are easy to spot:

11, 22, 33, 44, 55, 66, 77, 88 and 99

However, three (or more) digit numbers that are divisible by 11, such as 319 and 28347, are harder to spot.

One method for testing if a number is divisible by 11 involves first calculating its **alternating-digit sum**, as follows:

$$(1^{\text{st}} \text{ digit}) - (2^{\text{nd}} \text{ digit}) + (3^{\text{rd}} \text{ digit}) - (4^{\text{th}} \text{ digit}) + \dots$$

- The 1st digit is the one farthest to the left
- Digits are read from left to right
- Alternating means the signs change from - to +
- The first operation is always a subtraction, -

If the alternating-digit sum is a multiple of 11, then the original number is divisible by 11. The multiple of 11 can be:

- positive, e.g. 11, 22, ...
- negative, e.g. -11, -22, ...
- or zero, 0

For example, to test if 319 is divisible by 11:

$$3 - 1 + 9 = 11$$

This is a multiple of 11, so 319 is divisible by 11.

Complete the table below. Parts of the table have been completed for you.

Number	Alternating digits	Sum	Multiple of 11?
573	5-7+3	1	No
253	2-5+3	0	Yes
18095	1-8+0-9+5	-11	Yes
8184			
50021			
91949			
	1-9+0-9+1-6		

(2 marks)

- 2 (a)** One method for testing if a number is divisible by 11 involves first calculating its **alternating-digit sum**, as follows:

$$(1^{\text{st}} \text{ digit}) - (2^{\text{nd}} \text{ digit}) + (3^{\text{rd}} \text{ digit}) - (4^{\text{th}} \text{ digit}) + \dots$$

- The 1st digit is the one farthest to the left
- Digits are read from left to right
- Alternating means the signs change from - to +
- The first operation is always a subtraction, -

If the alternating-digit sum is a multiple of 11, then the original number is divisible by 11. The multiple of 11 can be:

- positive, e.g. 11, 22, ...

- negative, e.g. -11, -22, ...
- or zero, 0

Digits can only take the integer values 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9. A digit cannot be negative.

The five-digit number "**6371** x " is divisible by 11.

Use the test to find the value of the digit x .

(1 mark)

(b) The five-digit number "**24** ab **1**" is divisible by 11.

Show that the test gives the relationship

$$b = a - 1$$

(2 marks)

(c) The five-digit number "**24** ab **1**" from part (b) is divisible by both 11 and 3.

Find all the possible five-digit numbers that "**24** ab **1**" can be.

You should use a suitable test for divisibility by 3.

You may find the table below helpful.

a	$b = a - 1$	

(3 marks)

3 (a) One method for testing if a number is divisible by 11 involves first calculating its **alternating-digit sum**, as follows:

$$(1^{\text{st}} \text{ digit}) - (2^{\text{nd}} \text{ digit}) + (3^{\text{rd}} \text{ digit}) - (4^{\text{th}} \text{ digit}) + \dots$$

- The 1st digit is the one farthest to the left
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- Alternating means the signs change from - to +
- The first operation is always a subtraction, -

If the alternating-digit sum is a multiple of 11, then the original number is divisible by 11. The multiple of 11 can be:

- positive, e.g. 11, 22, ...
- negative, e.g. -11, -22, ...
- or zero, 0

A six-digit number in the form "**aaaaaa**" is **always** divisible by 11. This is because the test in question 1 gives

$$a - a + a - a + a - a = 0$$

Determine which out of the following six-digit numbers are **always** divisible by 11.

You may assume that $a \neq b \neq c$.

The example above is given in the first row.

Number	Working	Always divisible by 11?
"aaaaaa"	$a - a + a - a + a - a = 0$	Yes
"aabbcc"		
"ababab"		
"abbaaa"		
"abcabc"		

(2 marks)

- (b) The form "aaaabbbb" is an example of an eight-digit number that is always divisible by 11.

Using four digits of a and four digits of b , find two more examples of eight-digit numbers that are **always** divisible by 11.

Label the first digit as a .

(2 marks)

- 4 (a)** One method for testing if a number is divisible by 11 involves first calculating its **alternating-digit sum**, as follows:

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- The 1st digit is the one farthest to the left
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- Alternating means the signs change from - to +
- The first operation is always a subtraction, -

If the alternating-digit sum is a multiple of 11, then the original number is divisible by 11. The multiple of 11 can be:

- positive, e.g. 11, 22, ...
- negative, e.g. -11, -22, ...
- or zero, 0

A number is called palindromic if it is equal to itself when the digits are written in reverse order.

For example, the numbers 33, 181, 9999, 12321 and 579975 are all palindromic.

A palindromic number with an **even** number of digits is always divisible by 11.

Prove this statement is correct for an eight-digit palindromic number given by "*abcd dcba*".

(1 mark)

- (b)** A palindromic prime is a prime number that is also palindromic.

An example of a two-digit palindromic prime is 11.

An example of a five-digit palindromic prime is 11311.

The formula

$$P = n^2 + (n + 1)^2$$

is known to output palindromic primes for certain values of n .

Two such values are $n = 9$ and $n = 12$.

Use this information to find **two** three-digit palindromic primes.

(1 mark)

(c) Show that the number 111 is not a palindromic prime.

(1 mark)

(d) Using information from this question, explain clearly why palindromic primes, other than 11, cannot have an **even** number of digits.

(2 marks)

- 5 (a)** One method for testing if a number is divisible by 11 involves first calculating its **alternating-digit sum**, as follows:

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- The first operation is always a subtraction, -

If the alternating-digit sum is a multiple of 11, then the original number is divisible by 11. The multiple of 11 can be:

- positive, e.g. 11, 22, ...
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This divisibility rule can be proved using algebra.

This question will look at proving it for three-digit numbers.

The three-digit number, N , represented by the digits " abc " can be written algebraically using the formula

$$N = 100a + 10b + c$$

Show that this formula works for the number 234.

(1 mark)

- (b)** The formula in part (a) can be rearranged to

$$N = k(9a + b) + (a - b + c)$$

where k is a positive integer.

Find the value of the positive integer k .

(1 mark)

(c) The sum of two multiples of 11 is always divisible by 11.

Use this fact, together with the result in part (b), to prove that if the alternating-digit sum of the number " abc " is a multiple of 11, then N is divisible by 11.

(1 mark)

- 6 (a)** The method of factorisation in algebra can sometimes be used to test for divisibility by 11.

For example, the factorisation

$$x^2 - 1 = (x + 1)(x - 1)$$

can be used to show that 9999 is divisible by 11, by substituting $x = 100$ into both sides and simplifying:

$$\begin{aligned} 100^2 - 1 &= (100 + 1)(100 - 1) \\ 9999 &= 101 \times 99 \end{aligned}$$

Since one of the factors on the right-hand side, 99, is divisible by 11, the number on the left-hand side, 9999, must be divisible by 11.

Use algebra to show that

$$x^3 + 1 = (x + 1)(x^2 - x + 1)$$

(1 mark)

- (b)** By substituting $x = 10$ into

$$x^3 + 1 = (x + 1)(x^2 - x + 1)$$

show that 1001 is divisible by 11.

(1 mark)

- (c)** Use

$$x^3 + 1 = (x + 1)(x^2 - x + 1)$$

to show that 1729 is not divisible by 11.

Show your method clearly.

(3 marks)